

Customer Segmentation Using K-Means Clustering

EncoderX Remote Internship — Data Science | Week 03

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1. Executive Summary

This report presents a customer segmentation project completed as part of the EncoderX Remote Internship — Data Science, Week 03. The project applies K-Means clustering to the Mall Customer Segmentation Data dataset to identify groups of customers with similar income and spending behavior. After preprocessing, exploratory analysis, feature scaling, and evaluation using the Elbow Method and Silhouette Score, five customer clusters were selected. The final clustering solution produced a Silhouette Score of 0.5547 for 200 customers.

2. Project Objectives

- Understand and preprocess the customer dataset.
- Perform Exploratory Data Analysis (EDA).
- Select relevant features for clustering.
- Scale numerical features before applying K-Means.
- Determine a suitable number of clusters using the Elbow Method and Silhouette Score.
- Profile and interpret the resulting customer segments.
- Derive business insights from the identified customer groups.

3. Dataset

The project uses the Mall Customer Segmentation Data dataset available on Kaggle. The dataset contains 200 customers and includes CustomerID, Gender, Age, Annual Income (k\$), and Spending Score (1-100). CustomerID was treated as an identifier rather than a clustering feature.

Feature	Description
CustomerID	Unique customer identifier
Gender	Customer gender
Age	Customer age
Annual Income (k\$)	Annual income in thousands of dollars
Spending Score (1-100)	Customer spending behavior score

4. Technologies and Methods

Technologies: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

Methods: Data preprocessing, feature scaling, Exploratory Data Analysis, K-Means clustering, Elbow Method, Silhouette Score, PCA visualization, and cluster profiling.

5. Data Preprocessing

The data was examined for missing values and duplicates. CustomerID was excluded from clustering because it is an identifier. The final clustering features were Annual Income (k\$) and Spending Score (1-100). These numerical features were standardized using StandardScaler before K-Means was applied.

6. Exploratory Data Analysis

The analysis examined age, gender, annual income, spending score, age versus spending score, annual income versus spending score, and feature correlations. The relationship between annual income and spending score was the primary basis for the final two-feature clustering analysis.

7. K-Means Clustering

K-Means is an unsupervised clustering method that assigns observations to groups according to their distance from cluster centroids. Different values of K were evaluated from 2 through 10. The Elbow Method was used to inspect inertia, while Silhouette Score was used to assess the separation of clusters.

8. Cluster Evaluation

K	Silhouette Score
2	0.3213
3	0.4666
4	0.4939
5	0.5547
6	0.5399
7	0.5281
8	0.4552
9	0.4571
10	0.4432

Based on the project evaluation, K = 5 was selected for the final model. The final Silhouette Score was 0.5547.

9. Customer Segmentation Results

Cluster	Segment	Avg. Age	Avg. Income (k\$)	Avg. Spending	Customers
0	Standard Customers	42.72	55.30	49.52	81
1	High-Value Customers	32.69	86.54	82.13	39
2	Careful Spenders	25.27	25.73	79.36	22
3	Premium Low-Spending Customers	41.11	88.20	17.11	35
4	Young High-Spending Customers	45.22	26.30	20.91	23

10. Segment Interpretation

Standard Customers: The largest segment, with moderate average income and spending behavior.

High-Value Customers: Customers with relatively high average income and high spending activity.

Careful Spenders: Customers with relatively lower average income and a high spending score.

Premium Low-Spending Customers: Customers with relatively high average income but low spending activity.

Young High-Spending Customers: The segment profile reported in the project has relatively lower income and lower spending activity based on its recorded averages.

Segment names are descriptive labels assigned from the observed cluster averages.

11. Business Insights

- Targeted marketing can be tailored to the characteristics of each customer segment.
- Personalized offers can be designed around observed spending behavior.
- Higher-spending customer groups can be considered for loyalty and engagement programs.
- Customer segments can support more relevant product recommendations.
- Segment-level analysis can help guide customer retention and engagement activities.

12. Model Evaluation and Key Findings

The final model used five clusters and covered all 200 customers. The Silhouette Score of 0.5547 was the highest among the tested K values in the project table. The clustering was also examined through cluster visualizations and PCA visualization to understand the distribution of the customer groups.

13. Project Workflow

Dataset → Data Understanding → Data Cleaning → Data Preprocessing → EDA → Feature Selection → Feature Scaling → Elbow Method → Silhouette Score → K-Means Clustering → Cluster Visualization → Cluster Profiling → Segment Interpretation → Business Recommendations

14. Conclusion

This project demonstrates the use of unsupervised machine learning for customer segmentation using K-Means clustering. Annual Income and Spending Score were selected as the primary clustering features after preprocessing and scaling. Evaluation of multiple cluster counts resulted in a final five-cluster solution with a Silhouette Score of 0.5547. The resulting segments provide a structured view of differences in customer income and spending behavior and can be used to support targeted marketing, personalized engagement, product recommendations, and retention-oriented analysis.

15. Project Links

Resource	Link
GitHub Repository	https://github.com/gamepatt78/Customer_Segmentation
Dataset	https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python
Demo Video	https://drive.google.com/file/d/1GNWcJIBnLicvGdGSiRBX0PQj-wFeKccB/view?usp=sharing
LinkedIn Post	https://www.linkedin.com/posts/athwin-videsh-b40723125_encoderx-datascience-customersegmentation-activity-7507988758647066624-Kv4x

16. Author

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17. Internship Details

Program: EncoderX Remote Internship — Data Science

Batch: 02

Week: 03

Task: Customer Segmentation